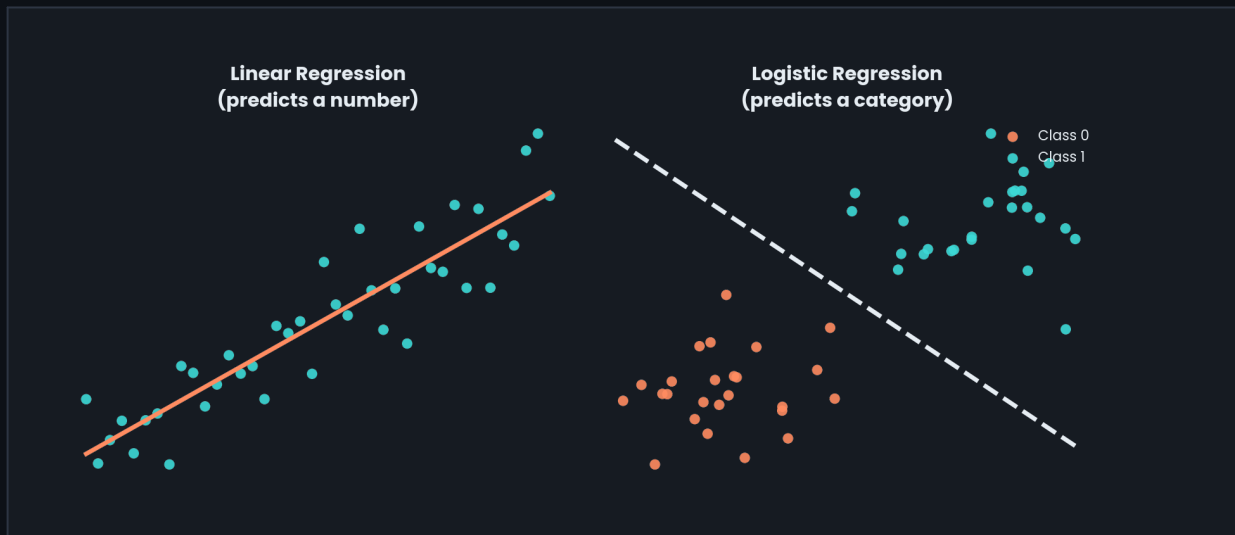


Linear & Logistic Regression

20 Practical Beginner Problems with Explanations & Visuals



10 Linear Regression problems • 10 Logistic Regression problems
Easy datasets • Clear explanations • Visual intuition

What's the Difference?

Both linear and logistic regression are supervised learning techniques — meaning they learn from labeled examples. The key difference is what they predict.

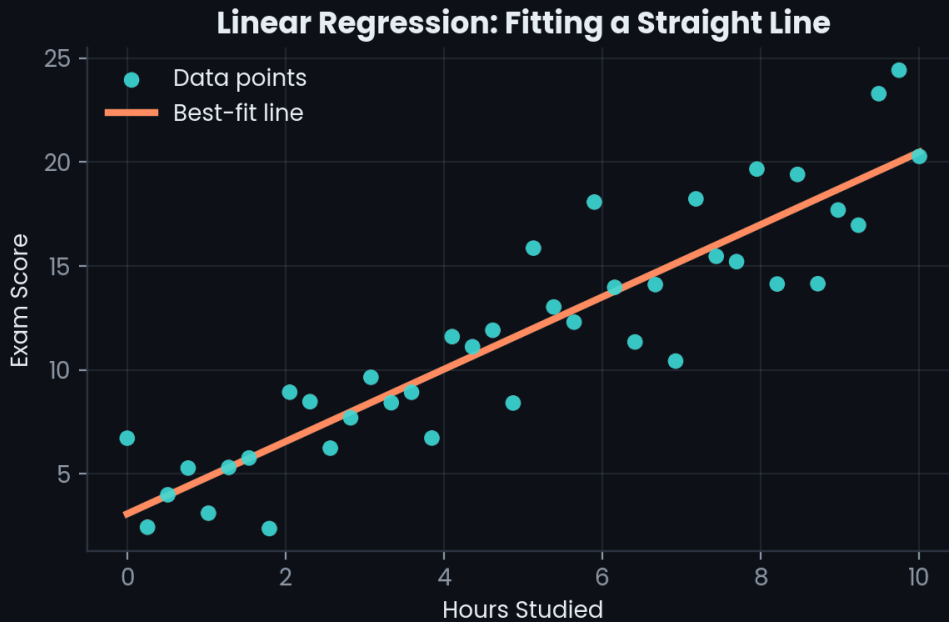
Linear Regression	Logistic Regression
Predicts a continuous number (price, score, temperature)	Predicts a category (yes/no, spam/not spam)
Fits a straight line through the data	Fits an S-shaped curve (sigmoid) that outputs a probability
Evaluated with R^2 , RMSE, MAE	Evaluated with Accuracy, Precision, Recall, Confusion Matrix

How to Use This Guide

For each problem below, follow the same five steps: load and explore the data, choose your features, split the data into training and test sets, fit the model, then check the relevant metric (R^2 or RMSE for regression; accuracy or a confusion matrix for classification). Most datasets mentioned here (Titanic, 50_Startups, Diabetes, Heart Disease UCI, Telco Churn) are freely available on Kaggle and small enough to run in a few minutes.

LINEAR REGRESSION

Linear regression predicts a continuous number by fitting the straight line that best follows the pattern in your data — minimizing the distance between the line and every data point.



1

House Price Prediction

Predict the selling price of a house from square footage, number of bedrooms, and location. Price tends to rise roughly proportionally with these features, which is exactly what a straight-line model captures.

2

Student Score Prediction

Predict a student's exam score from hours studied. A classic single-variable example showing a clear linear trend between study time and performance.

3

Salary Estimation

Predict an employee's salary from years of experience. Useful for HR analytics since experience and salary often grow together in a fairly linear way.

4

Sales Forecasting

Predict next month's sales from advertising spend across TV, radio, and online channels. Helps a business see which channel gives the best return.

5**Car Mileage Prediction**

Predict a car's fuel efficiency (mpg) from engine size, weight, and horsepower. A common example in the automotive and engineering domain.

6**Temperature Prediction**

Predict tomorrow's temperature from today's temperature, humidity, and wind speed. A gentle introduction to time-based regression without full time-series models.

7**Crop Yield Prediction**

Predict the yield of a crop (tons per acre) from rainfall, fertilizer used, and temperature. Common in agricultural data science.

8**Electricity Bill Prediction**

Predict the monthly electricity bill from number of appliances, hours used, and household size. Practical for personal finance or smart-home apps.

9**Startup Profit Prediction**

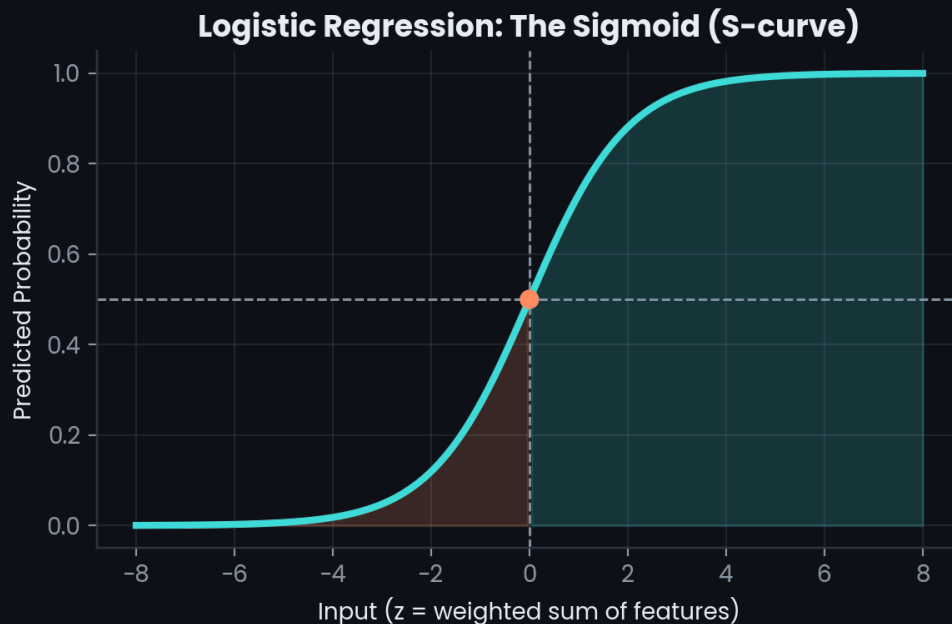
Predict a startup's profit from R&D; spend, administration spend, and marketing spend. A favorite introductory dataset for multiple linear regression.

10**Used Car Price Prediction**

Predict the resale price of a used car from age, mileage, brand, and condition score. Great practice mixing numeric and categorical features.

LOGISTIC REGRESSION

Logistic regression predicts a category (usually yes/no) by squeezing a weighted sum of the inputs through an S-shaped sigmoid curve, turning it into a probability between 0 and 1.



1

Email Spam Detection

Classify an email as spam or not spam based on word frequency features. The classic binary classification starter problem.

2

Disease Diagnosis

Predict whether a patient has a disease (yes/no) from test results like blood pressure, glucose level, and BMI. Common in healthcare ML, e.g. diabetes prediction.

3

Customer Churn Prediction

Predict whether a customer will leave a subscription service based on usage patterns and complaints. Important for retention strategy.

4

Loan Approval Prediction

Predict whether a loan application will be approved or rejected based on income, credit score, and existing debt. Common in fintech and banking.

5**Titanic Survival Prediction**

Predict whether a passenger survived the Titanic disaster based on age, gender, and ticket class. The most famous beginner logistic regression dataset.

6**Exam Pass/Fail Prediction**

Predict whether a student passes or fails based on hours studied and attendance percentage. Simple two-feature case, easy to visualize.

7**Click-Through Prediction**

Predict whether a user will click an online ad based on age, browsing time, and device type. Used heavily in digital marketing and ad-tech.

8**Heart Disease Risk Prediction**

Predict whether someone is at risk of heart disease based on cholesterol, age, and blood pressure. Another strong healthcare classification case.

9**Employee Attrition Prediction**

Predict whether an employee will leave the company based on satisfaction score, salary, and work hours. A common HR analytics problem.

10**Fraud Detection**

Predict whether a credit card transaction is fraudulent based on amount, location, and time. Introduces classification on imbalanced data, a key real-world challenge.

The 5-Step Workflow (for every problem)

1. Load & Explore	Read the dataset, check for missing values, and look at summary statistics or a quick plot.
2. Choose Features	Pick the input columns (X) that logically relate to what you're predicting (y).
3. Train/Test Split	Hold back ~20–30% of the data to test the model on examples it hasn't seen.
4. Fit the Model	Use scikit-learn's <code>LinearRegression()</code> or <code>LogisticRegression()</code> and call <code>.fit(X_train, y_train)</code> .
5. Evaluate	Regression: check R^2 and RMSE. Classification: check accuracy and the confusion matrix.

Where to Find Free Datasets

Kaggle hosts small, beginner-friendly versions of most datasets referenced in this guide, including Titanic, 50_Startups, the Pima Diabetes dataset, Heart Disease UCI, and Telco Customer Churn. Each is small enough to load and train a model on in just a few minutes.

Practice tip: implement one linear and one logistic problem fully end-to-end before moving to the next — depth beats breadth when you're starting out.